Elective 7-I (Selective)

SWE 4731 **Advanced Network Protocols** **Credit: 3.00**

Review of networking protocols; Router and Switch architectures, Packet Classification, Packet scheduling and fair queuing, Protocol Processing; Overview of Linux Network kernel programming; Network Congestion control, Data Centre Networking, Traffic analysis, Software Defined Networks.

Recommended Texts:

1. G. Varghese and J. Xu, *Network Algorithmics: An Interdisciplinary Approach to Designing Fast Networked Devices*, 2nd ed. Morgan Kaufmann, 2022
2. C. Benvenuti, *Understanding Linux Network Internals*, 1st ed. O'Reilly Media, Inc., 2006
3. K. R. Fall and W. R. Stevens, *TCP/IP Illustrated*, 2nd ed. Addison-Wesley Professional, 2011, vol. I, II, III

SWE 4739 **Embedded Software Development** **Credit: 3.00**

This course covers computing elements, structures in embedded software, resource access protocols, uniprocessor scheduling, programming-language support, languages for model-driven development, worst-case execution time analysis, and overview of embedded distributed systems. Other topics include specification and design of embedded systems, specification languages, hardware/software co-design, performance estimation, co-simulation, embedded architectures, processor architectures and software synthesis, system-on-a-chip paradigm, retargetable code generation and optimization, verification and validation, environmental issues and considerations.

SWE 4741 **Computer and Information Security** **Credit: 3.00**

Computer and data security goals, Challenges of protecting computer and information.

Access Control: Access control models, Discretionary Access Control (DAC), Mandatory Access Control (MAC), Commercial Security Policies, Role-based access control, Credentials and Delegation, dynamic access control.

Threats and Attacks landscape: Types and examples of computer and information security threats; Methods of attack; malware, social engineering and TCP/IP based attacks;

Vulnerabilities: Design and application vulnerabilities, Common vulnerability exposure;

Attackers: Attackers type, skill and motivation;

Information Flow: Program analysis techniques, Quantitative information flow, Covert channels;

Perimeter and host Defenses: Perimeter attacks, security zones and devices, configuring a DMZ, NAT router, VPNs, protections against web threats, Malware protection, password attacks, hardening a Windows system, managing file system security, security of a Linux system;

Data Defenses: Redundancy through RAID, proper management of backups and restores, file encryption, implementing secure protocols, and cloud computing;

Web security: Basic concepts of securing web applications, fortifying the internet browser, securing e-mail from e-mail attacks, security considerations about networking software. Internet security controls;

Law and Ethics: Digital Security Act, Data protection, Copyright, Trade Mark and Patent, Computer misuse act, Ethics in computer security.

Recommended Texts:

1. C. P. Pfleeger et al., *Security in Computing*, 6th ed. Addison-Wesley Professional, 2023
2. R. Anderson, *Security Engineering: A Guide to Building Dependable Distributed Systems*, 3rd ed. Wiley, 2020

SWE 4737 **Computer Animation** **Credit: 3.00**

Introduction to computer animation, Technical background for computer animation, Technical background for computer animation, Introduction to computer animation software, Interpolation and Basic Techniques, Skeletal animation (Motion capture), Skeletal animation (Motion capture), Physically based animation, Group behavior and crowed animation, Fluids: Clouds, Water, Fire; Figures, Facial, and Behavior.

Recommended Texts:

1. R. Parent, *Computer Animation: Algorithms and Techniques*, 3rd ed. Morgan-Kaufmann, 2012
2. M. Orourke, *Principles of Three-Dimensional Computer Animation*, 3rd ed. W. W. Norton & Company, 2003
3. L. Pocock and J. Rosebush, *The Computer Animator's Technical Handbook*, 1st ed. Morgan Kaufmann, 2001

SWE 4732 **Advanced Network Protocols Lab** **Credit: 0.75**

Sessional works based on SWE 4731.